Track Me Live — Functional Specification Document

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1. Overview

Track Me Live lets a person share their real-time GPS location with a trusted contact for the duration of a phone call, or manually at any time, without requiring the recipient to install the app. The location appears as a customizable animated character (mascot) on a live map link.

Problem it solves: existing live-location tools (WhatsApp, Messenger, Google Maps) require opening a separate sharing flow inside a messaging app. None tightly couple sharing to the act of being on a phone call, and none offer a personalized, non-generic tracking marker.

Target users: people walking alone at night, commuters checking in with family, solo travelers, and parents/partners who want a lightweight, low-friction safety check-in tool.

One-line pitch: "Share where you are, the moment you're on the phone — with a character that feels like you."

2. Scope

In v1	Deferred to v2+
Call-triggered sharing prompt	ETA calculation
Manual sharing (not tied to a call)	SOS / panic button
Live web-link map, no install required	Share history log
Auto-expiry on call end or timeout	Trusted contacts list for one-tap sharing outside calls
Mascot picker (4-6 base characters)	Expanded mascot marketplace / custom uploads
Stop sharing anytime	Multi-recipient broadcast beyond the call participant
Notification when recipient opens the link	

3. User roles

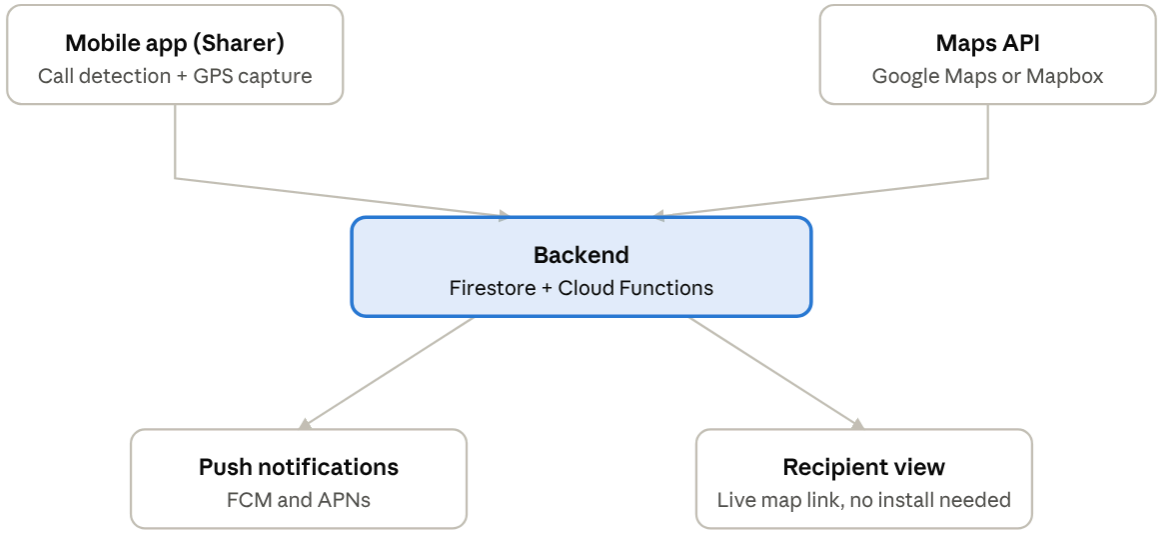
Role	Can do
Sharer	Start/stop sharing, pick mascot, choose precise vs. approximate location, see when the link was viewed
Viewer (recipient)	Open the live map link, see the sharer's position update in real time, no account or app install required

4. Functional requirements

Feature	Trigger	Behavior	Edge case handling
Call-triggered sharing	Active call detected (CallKit / Telecom API)	One-tap banner offers "Share live location"	If declined, no location data leaves the device
Manual sharing	User taps share from home screen	Starts a session without needing a call	Can run alongside a call-triggered session
Live map link	Session starts	Generates a unique, expiring URL opened in any browser	Link invalidated once session ends
Auto-expiry	Call ends, or timer set by user	Session status flips to stopped; last location frozen	Default max duration 1 hour even if not manually stopped
Mascot marker	Live map loads	Animated character marks the sharer's position	Falls back to a default neutral marker if no mascot chosen
Stop sharing	User taps Stop	Session ends immediately, pings stop writing	Viewer sees "sharing ended" state, not a frozen live indicator
View notification	Recipient opens link	Sharer gets a push notification	Only fires once per link open, not per refresh
Battery-aware pings	Ongoing session	GPS write frequency drops as battery falls below set thresholds	Never drops below 1 ping per 60s while active

5. System architecture

Mobile app streams location through the backend to the recipient view



system architecture · client, backend, delivery

The mobile app detects an active call and captures GPS, writing pings to the backend (Firestore for live data, Cloud Functions for session logic, expiry, and security rules). The Maps API renders both the backend's reverse-geocoding and the recipient's live map. The backend fans out to push notifications (call-end alerts, view-confirmation) and to the recipient view — a web page requiring no install, reading directly from the live session document.

6. Database schema

Firestore (document-based), chosen for built-in real-time listeners over a self-built WebSocket layer.

Collection	Key fields	Notes
users/{userId}	phone, name, mascot_id (FK)	Profile and mascot selection
users/{userId}/trustedContacts/{id}	contact_user_id, relation	Subcollection, scoped per user

Collection	Key fields	Notes
shareSessions/{sessionId}	user_id, recipient_id, status, started_at, expires_at	One doc per active/past share
shareSessions/{sessionId}/pings/{id}	lat, lng, recorded_at	Subcollection the recipient's map listens to live
mascots/{id}	character, color	Base character library
callLogs/{id}	user_id, session_id, call_time	Links a call to the session it triggered

Design decisions: ping subcollections auto-delete 24h after session end via a scheduled Cloud Function; GPS writes throttled to every 3-5 seconds; `expires_at` auto-stops sharing even on app crash; Firestore security rules restrict a recipient to reading only their own session's pings.

7. Non-functional requirements

Area	Requirement
Performance	Live map reflects a new ping within 5 seconds of capture
Battery	Ping frequency scales down below 20% battery; never below 1 ping/60s while active
Security	All session links are single-use tokens with expiry; location data encrypted in transit and at rest
Offline behavior	If the sharer loses signal, the viewer sees "last seen" timestamp rather than a frozen live marker
Accessibility	VoiceOver labels on map and mascot; Dynamic Type support; status never conveyed by color alone

8. Permissions and data handling

- **Location**: requested on first share attempt, not at install; "while using" is sufficient except for call-triggered background sharing, which needs "always allow" with a clear in-app explanation.
- **Contacts**: optional, only needed for the trusted-contacts list (v2 feature).
- **Notifications**: requested to enable view-confirmation and session-expiry alerts.
- **Retention**: ping data auto-deleted 24 hours after a session ends; session metadata (not location) retained for 30 days for support/debugging, then purged.
- **Deletion**: users can delete their account and all associated data at any time from Settings.

9. Error and edge cases

Scenario	Handling
No GPS signal	Viewer sees "last seen" timestamp and location instead of a stale live marker
Call drops mid-share	Session continues per the user's chosen duration, not tied strictly to call state
Recipient has no app	Link opens a browser-based map view, no install required
App killed in background	expires_at field server-side force-stops the session even without a client heartbeat
Recipient opens link after expiry	Shows "this location is no longer being shared" state, not a stale pin

10. Competitive landscape

App	Call-triggered	Custom tracking icon	Install required for recipient
Mapfia (historical)	Yes	No	Yes
WhatsApp Live Location	No	No	Yes
Messenger location	No	No	Yes

App	Call-triggered	Custom tracking icon	Install required for recipient
Life360	No	No	Yes
bSafe	No	No	Yes
Noonlight	No	No	Yes
Strava Beacon	No	No	No
Track Me Live	**Yes**	**Yes**	**No**

No major player combines a call-trigger with a no-install recipient link and a personalized marker — this is the identified gap.

11. UI and design system overview

Screens: Onboarding, Home/dashboard, Active-call overlay, Live tracking screen, Mascot picker, Contacts, Settings, Share history.

Mascot system: 4-6 base characters (fox, ghost, robot, astronaut), customizable by color and accessory; animates subtly (idle bounce) on the live map as the position marker; falls back to a default neutral marker if unset.

Design tokens: SF Pro typography (iOS HIG), SF Symbols for system icons, light/dark mode, calm primary palette with a distinct alert accent for SOS/stop actions. Full screen-by-screen visual spec to follow as a separate design pass.

12. Success metrics and glossary

Success metrics: time-to-share under 3 taps; live map load time under 2 seconds; under 5% of sessions failing to auto-expire correctly.

Glossary

Term	Meaning
Sharer	The user broadcasting their live location
Viewer	The recipient watching the live map link
Session	One instance of active location sharing, from start to expiry
Ping	A single GPS coordinate write to the active session

Term	Meaning
Mascot	The customizable character used as the live-location marker